Video RAG Retrieval Project

https://github.com/aiyshwary/Video-RAG-Retrieval

Python FAISS OpenCLIP ViT-GPT2 Sentence Transformers RAG

OVERVIEW

A semantic video search system that extracts frames, generates scene captions using ViT-GPT2, and embeds them with SentenceTransformers and OpenCLIP into a dual FAISS index, enabling natural language queries to retrieve the most relevant video scenes with timestamps.

IMPACT

Built a Visual RAG system that makes unstructured video content semantically searchable: a text query retrieves the most relevant scenes with timestamps by matching against dual text and image embedding indexes via cosine similarity.

KEY DETAILS

- Extracts frames at a configurable FPS and generates natural-language scene captions using ViT-GPT2 image captioning.
- Computes text embeddings with SentenceTransformers (MiniLM) and image embeddings with OpenCLIP for dual-modality indexing.
- Stores embeddings and metadata in two parallel FAISS indexes supporting independent text or image-based retrieval.
- Query interface accepts a natural language string and returns top-k matching scenes with frame references and timestamps.
- Modular architecture (preprocessing, embeddings, storage, retrieval) designed for extension to production vector databases.